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Wednesday

Lecture 7 – Support Vector Machine

INT104 Artificial Intelligence

Support Vector Machine

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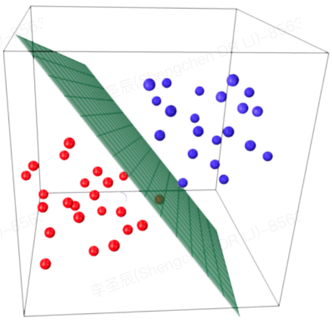
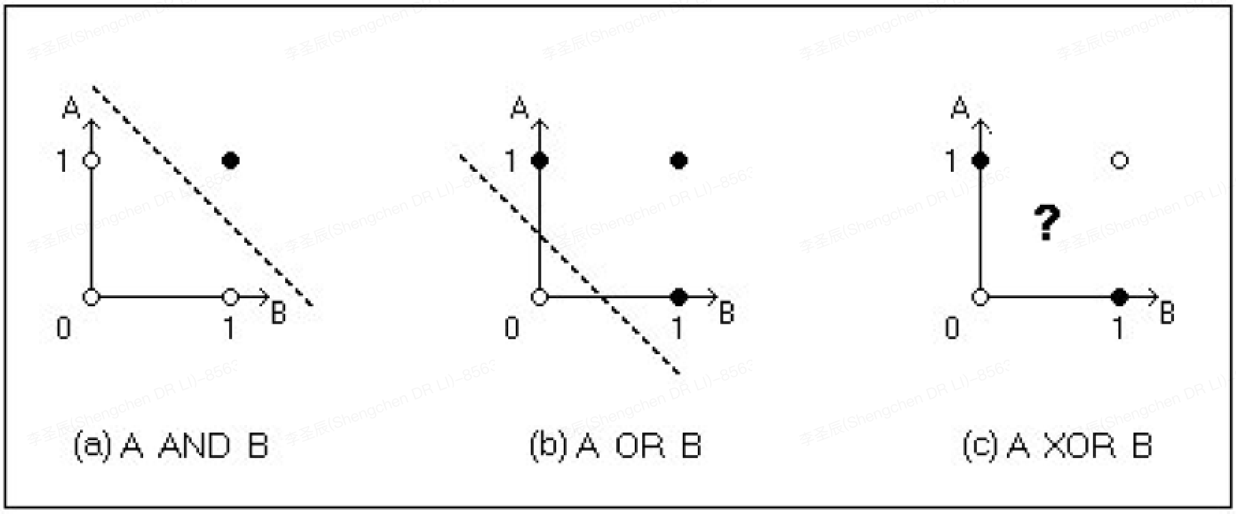
Linearly Saparable

- *Ideally, if the feature resulted is good enough, a simple linear function would solve the problem of classification.*
- *But the relationship between samples is not necessary to be a linear relationship.*



Logic Function as Linear Combination

- We know consider the most basic logic functions: AND & OR & XOR
- How can we find the proper linear discriminant function?



Large Margin Classification

- If classes are linearly separable, two classes can be separated with a straight line
- SVM pursues the widest possible street between classes

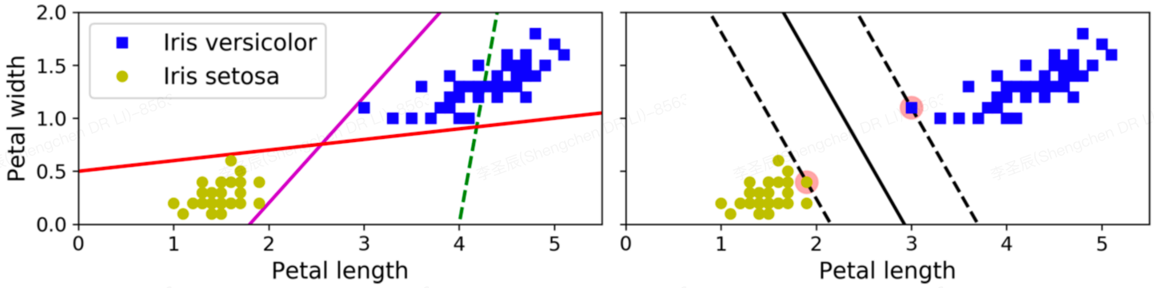


Figure 5-1. Large margin classification

Motivation: Many Separating Hyperplanes, Which One to Choose?

Assume a binary classification dataset:

- Training data: $\{(x_i, y_i)\}_{i=1}^n$
- Feature vector: $x_i \in \mathbb{R}^d$
- Label: $y_i \in \{+1, -1\}$

A linear decision boundary in $\mathbb{R}^d$ is a hyperplane:

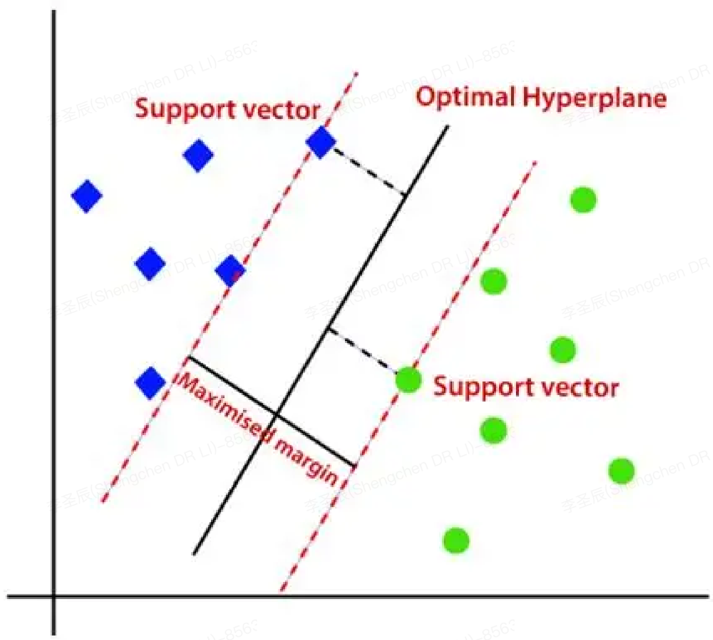
Prediction rule:

$$\hat{y} = \mathrm{sign}(w^\top x + b)$$

Correct classification condition for a sample (x_i, y_i) :

$$y_i(w^\top x_i + b) > 0$$

Support Vector



Distance to a Hyperplane (Geometric Quantity)

Consider the hyperplane $\mathcal{H} = \{x : w^\top x + b = 0\}$.

The **signed distance** from a point x to $\mathcal{H}$ is:

$$\mathrm{dist}_\pm(x, \mathcal{H}) = \frac{w^\top x + b}{\|w\|}$$

- $\|w\|$ is the Euclidean norm of w
- The sign indicates which side of the hyperplane the point lies on

For a labeled sample (x_i, y_i) , define the **geometric margin**:

$$\gamma_i = \frac{y_i(w^\top x_i + b)}{\|w\|}$$

The margin of the classifier (w, b) on the dataset:

$$\gamma = \min_i \gamma_i$$

Lagrange Multiplier

- Optimisation: Largest Street (margin)
- Constrains: Saparable Hyper–plane

Mathematically the objective of optimisation with contrains (essentially a Lagrange Multiplier) is:

$$L(w, b, \alpha) = \frac{1}{2}\|w\|^2 + \sum_{i=1}^{n} \alpha_i (1 - y_i(w^\top x_i + b))$$

Feature Scales

- The decision boundary could be much better if the feature is properly scaled.

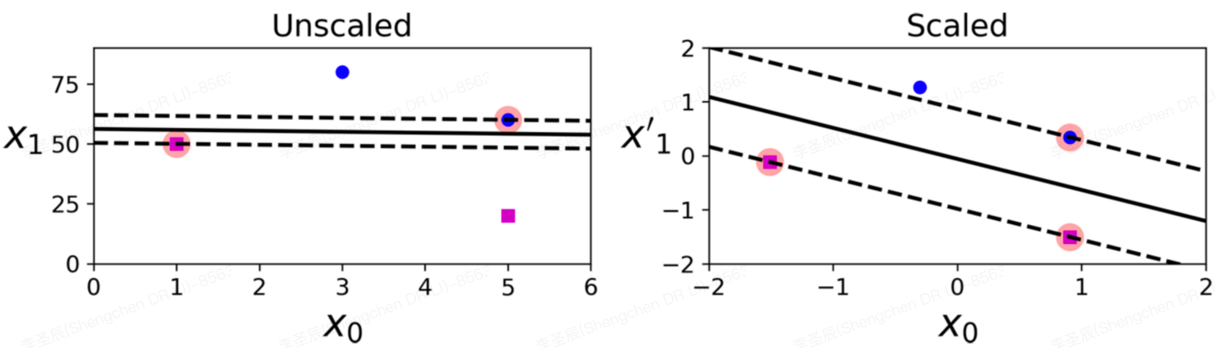


Figure 5-2. Sensitivity to feature scales

Hard Margin Classification

- All instances being off the street and on the right side is named “hard margin classification”
- The main limitation of hard margin classification is
  - The data must be linearly separable
  - There are little outliers

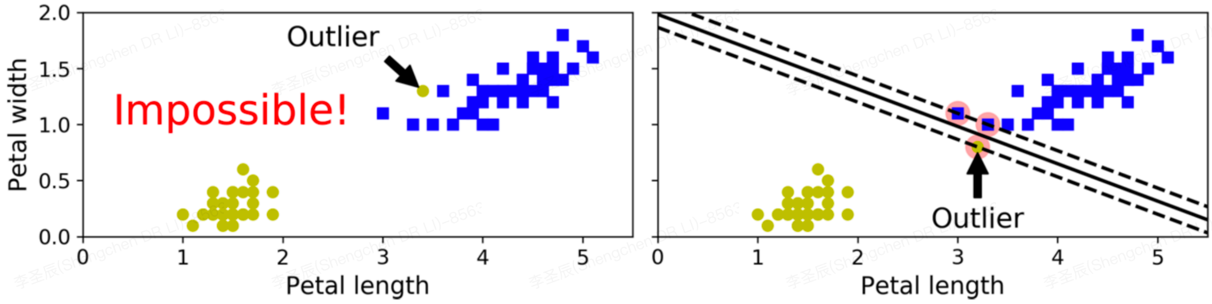


Figure 5-3. Hard margin sensitivity to outliers

Soft Margin Classification

- Soft Margin Classification provides more flexibility
- The algorithm balances
  - The width of street
  - The amount of margin violations
- A hyper–parameter C is defined
  - A low value of C leads to more margin violation
  - A high value of C limits the flexibility

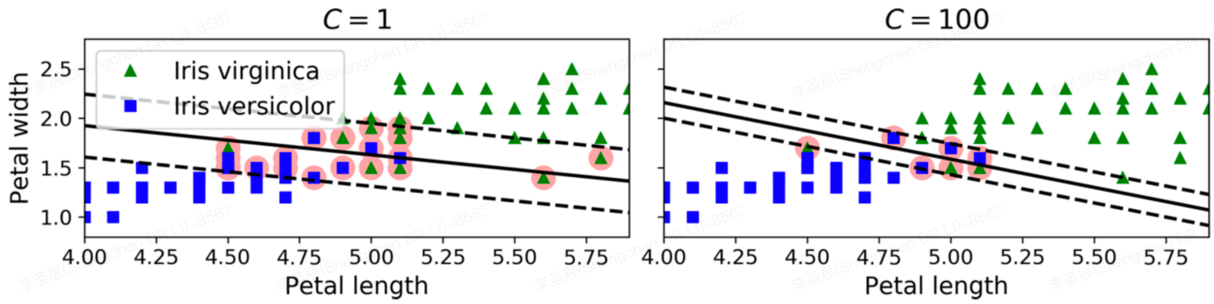


Figure 5-4. Large margin (left) versus fewer margin violations (right)

Polynomial Feature

- A second feature could be added such as $x_2 = (x_1)^2$

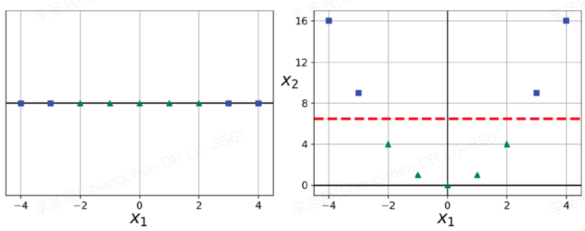


Figure 5-5. Adding features to make a dataset linearly separable

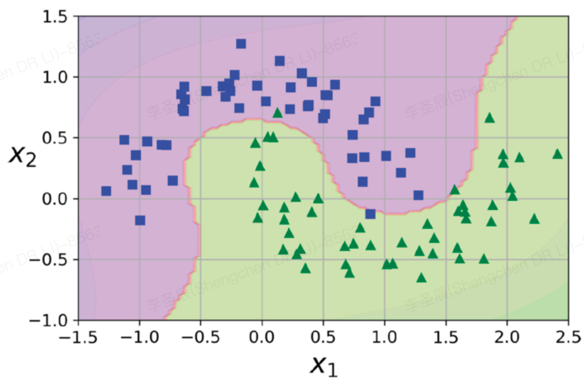


Figure 5-6. Linear SVM classifier using polynomial features

Polynomial Kernel

- A kernel can be used instead of adding the polynomial features

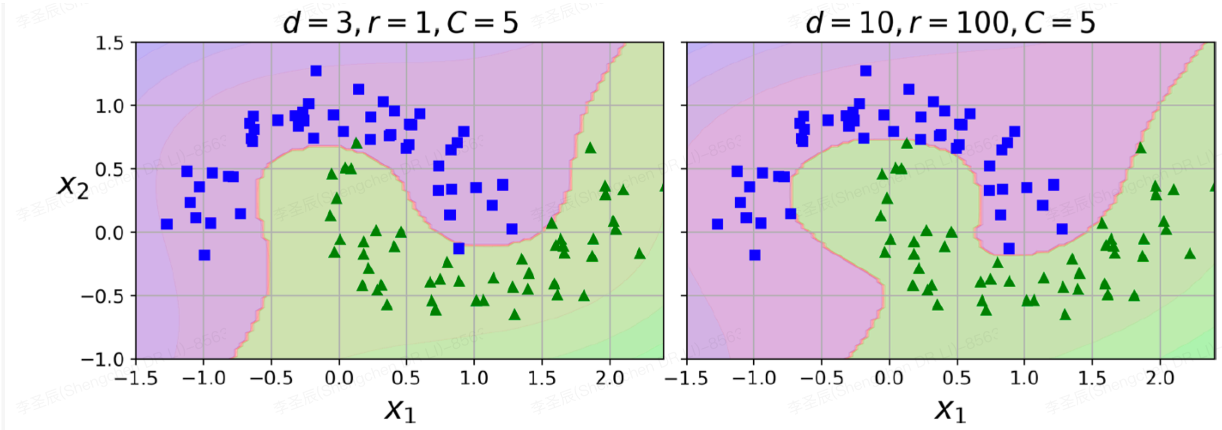


Figure 5-7. SVM classifiers with a polynomial kernel

Similarity Feature

- Samples could be clustered with a landmark (reference point)
- The similarity function could be defined as a Gaussian Radial Basis Function

$$\phi_\gamma(\mathbf{x}, l) = e^{-\gamma||\mathbf{x}-l||^2}$$

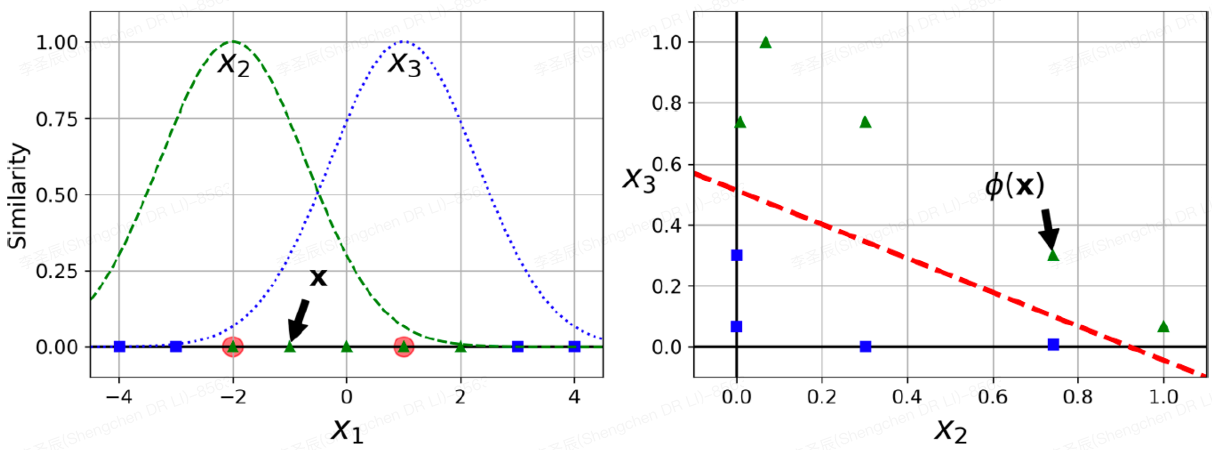


Figure 5-8. Similarity features using the Gaussian RBF

Gaussian RBF Kernel

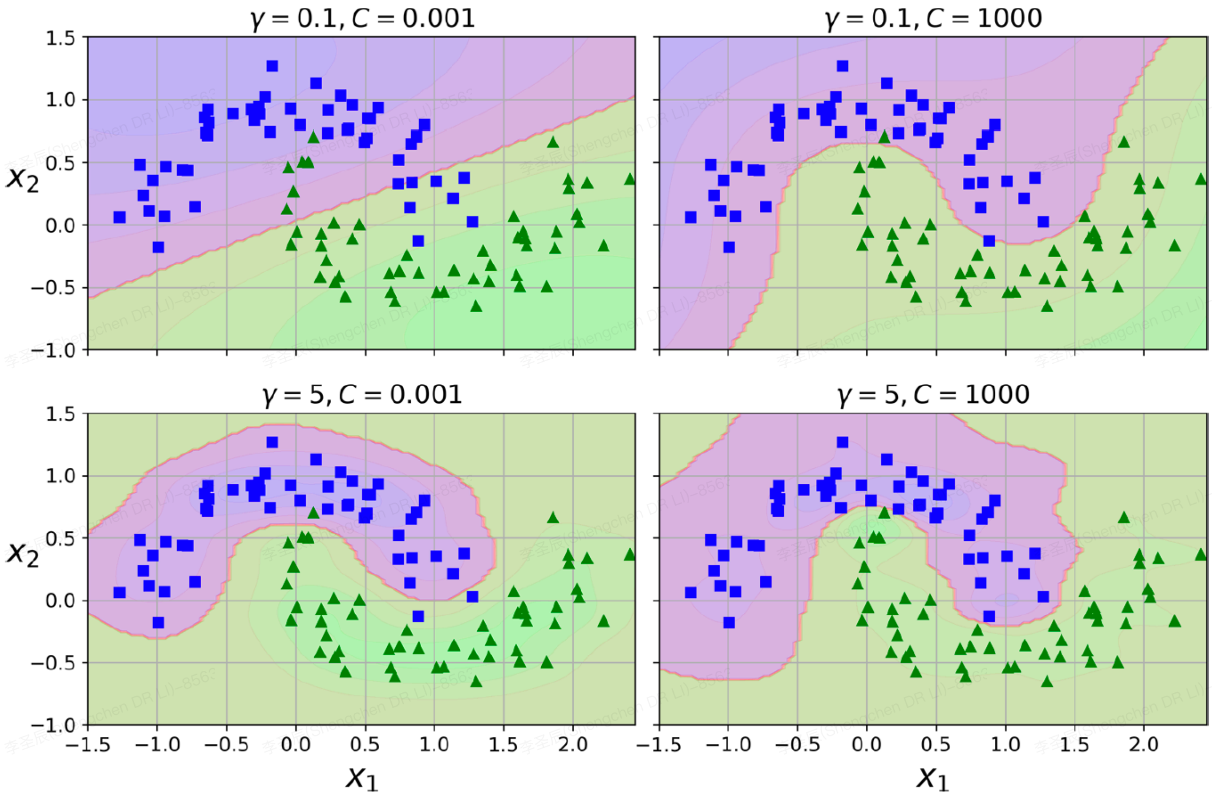
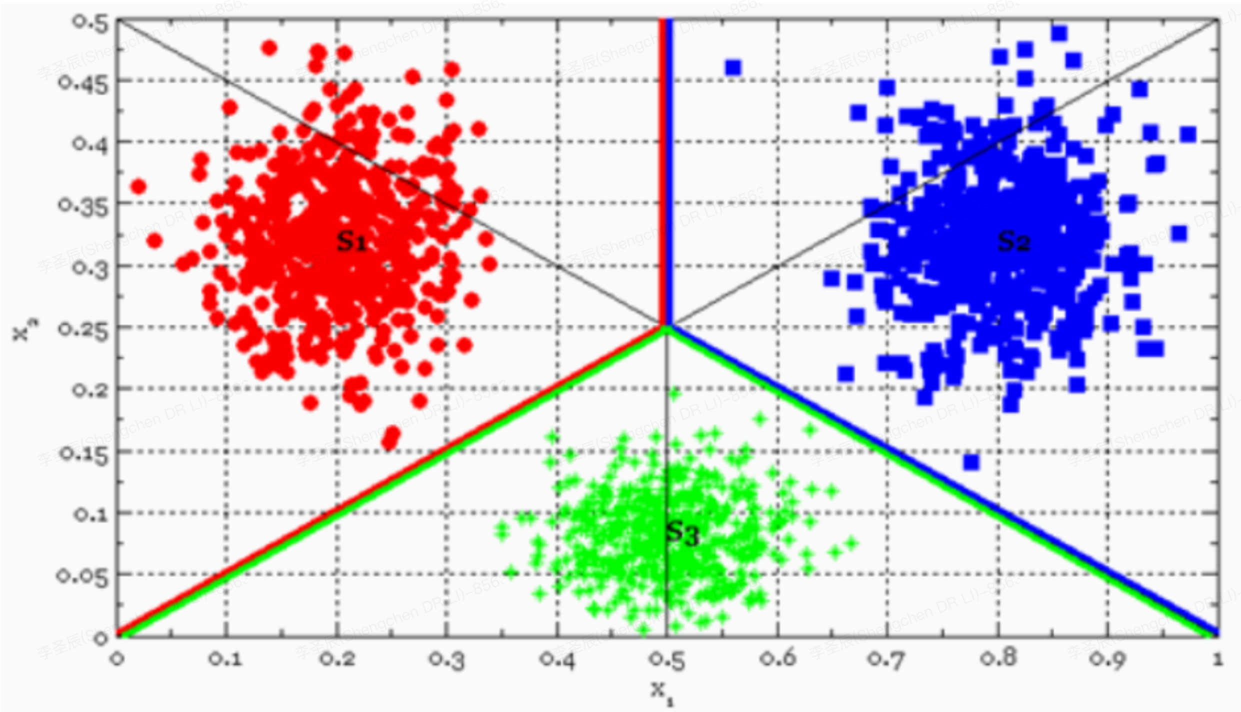
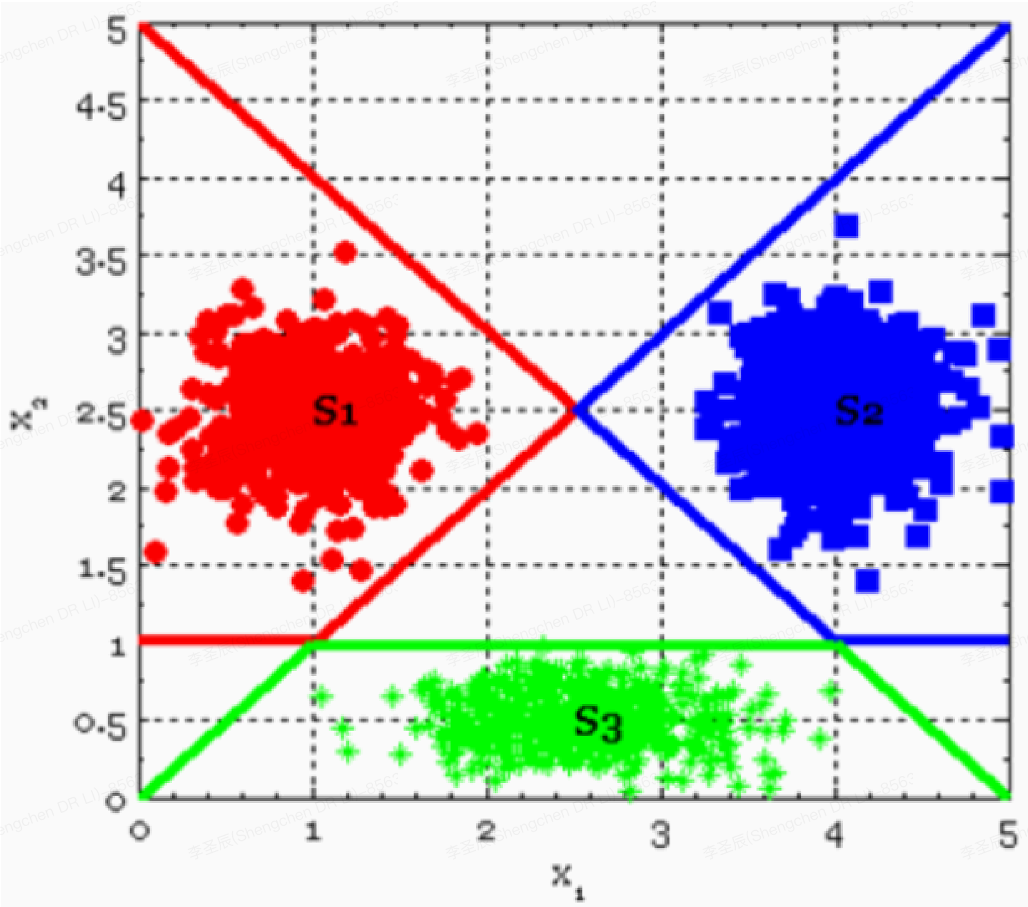


Figure 5-9. SVM classifiers using an RBF kernel

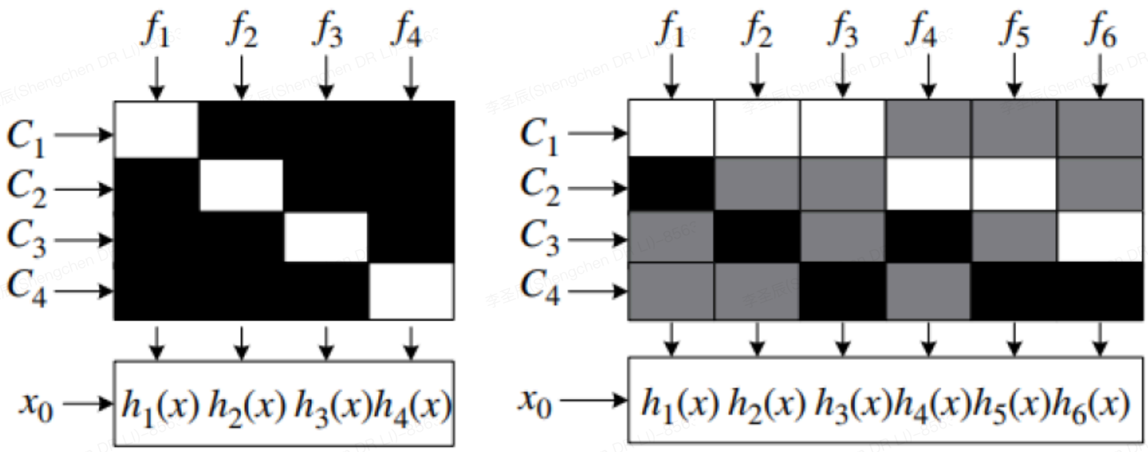
Multi-class Classification



Multi-class Classification



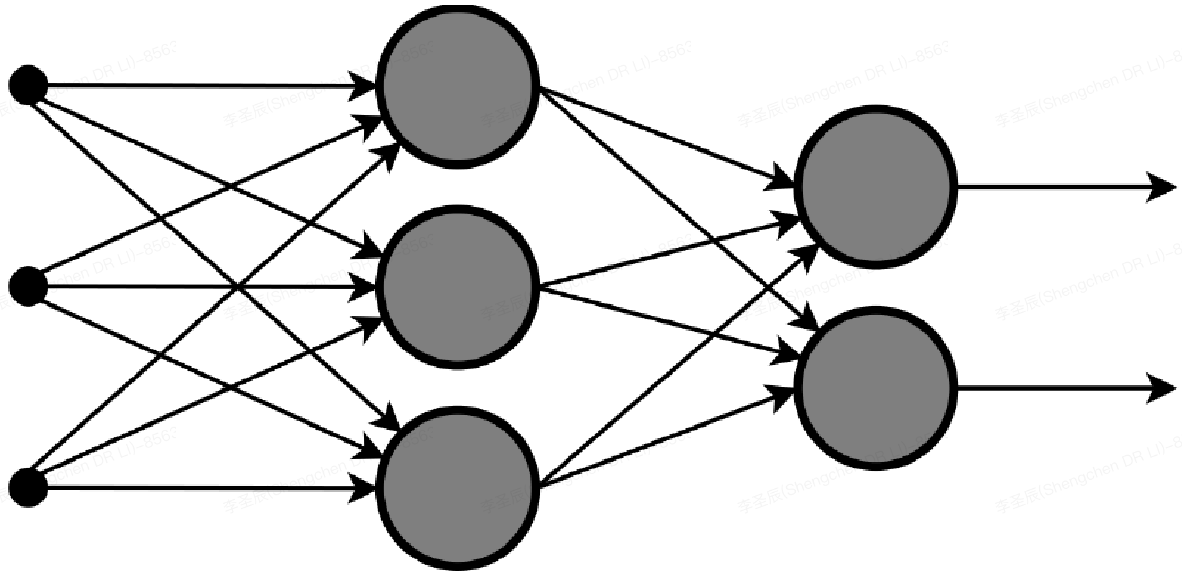
Multi-class Classification



Please refer to *Zhou2016* and `https://iep.utm.edu/prop-log/`

Perceptrons

- The boundaries between classes are not necessary linear but can be approximate as a combination of linear functions.
- Could be single layer or multiple layer



- There is a threshold process after the output of each neuron, which is named as activation function

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Deep Learning

- Perceptron is a basic unit of Neural Network.
- Neural Networks with multiple layers are considered as deep learning systems.
- The extension along space axis results Convolutional Neural Network (with some simplication)
- The extension along time axis results Recurrent Neural Network (with more controls between time slices)

SVM Regression

- Fit as many instances as possible on the street
- Limit margin violations
- Width of street is controlled by a hyper-parameter ϵ

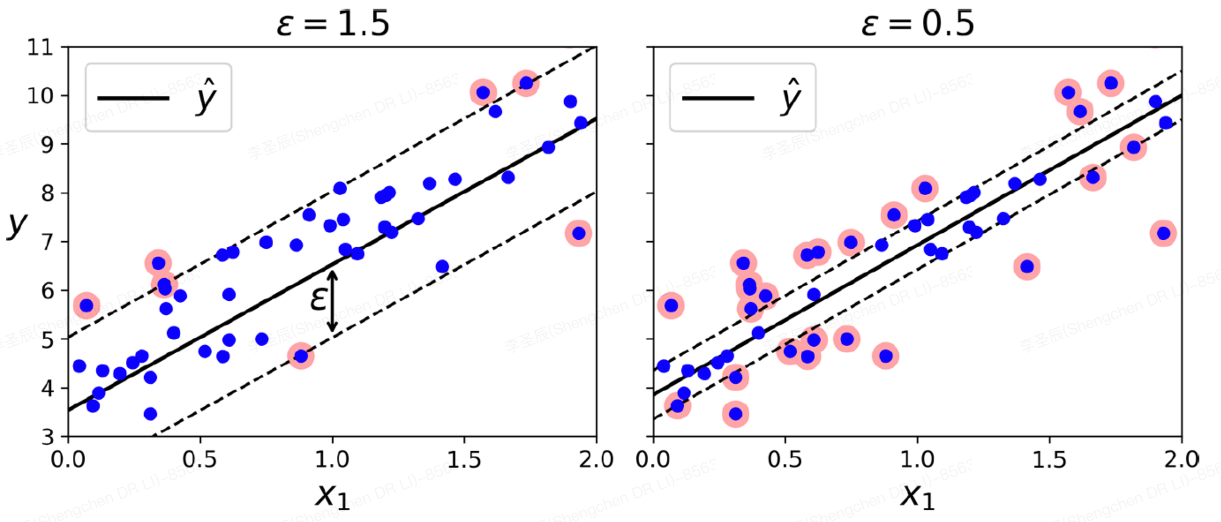

Figure 5-10. SVM Regression

SVM Regression

- Kernel tricks could also be used

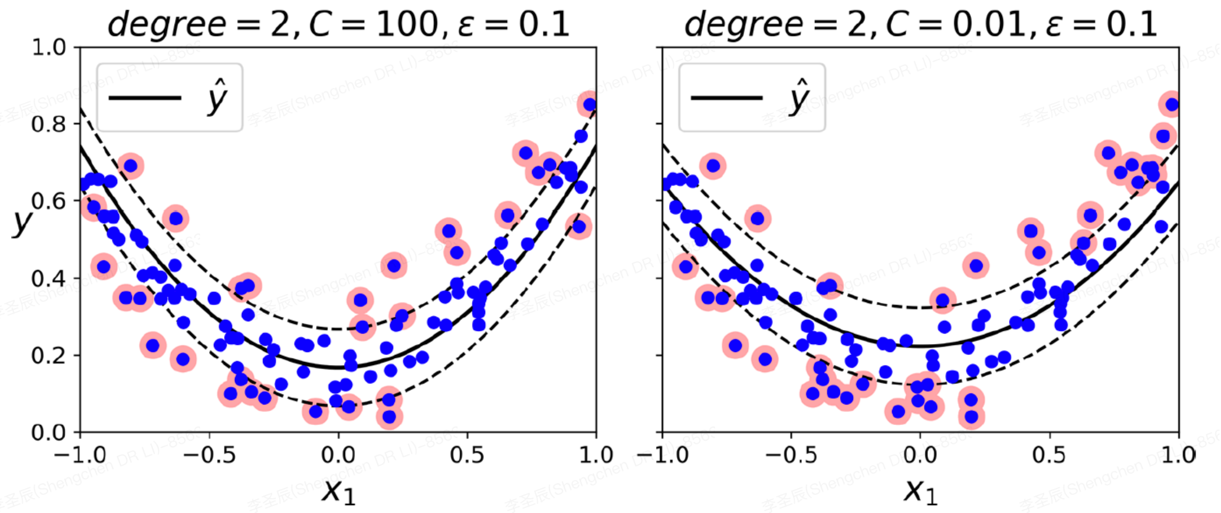

Figure 5-11. SVM Regression using a second-degree polynomial kernel